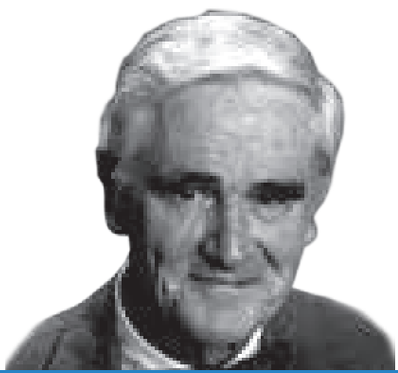
UNIT 13

ORGANIC NITROGEN COMPOUNDS



Donald James Cram

Donald James Cram was an American chemist who shared the 1987 Nobel Prize in Chemistry with Jean-Marie Lehn and Charles J. Pedersen "for their development and use of molecules with structure-specific interactions of high selectivity." They were the founders of the field of host–guest chemistry Cram expanded upon Charles Pedersen's ground-breaking synthesis of crown ethers, two-dimensional organic compounds that are able to recognize and selectively combine with the ions of certain metal elements. He also did work in stereochemistry and Cram's rule of asymmetric induction is named after him.

U3U7L5

Learning Objectives

After studying this unit the student will be able to

- understand isomerism in organic nitro compounds
- describe the preparation and properties of nitro compounds
- classify amines as primary, secondary and tertiary
- describe the methods of preparation of amines
- explain the properties of amines
- distinguish between primary, secondary and tertiary amines
- describe the method of preparation of diazonium salts
- explain the preparation and properties of cyanides

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